

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Construction Technology, Grade 11

Workplace Preparation

TCJ3E

This course enables students to develop technical knowledge and skills related to carpentry, masonry, electrical systems, heating and cooling, and plumbing for residential construction. Students will gain hands-on experience using a variety of materials, processes, tools, and equipment to design, lay out, and build projects. They will create and read technical drawings, learn construction terminology, interpret building codes and regulations, and apply mathematical skills as they develop construction projects. Students will also develop an awareness of environmental and societal issues related to construction technology, and will explore postsecondary and career opportunities in the field.

Prerequisite: None

A. CONSTRUCTION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** identify and describe a variety of construction materials, components, and processes;
- A2.** describe the scope and purpose of building codes, and identify other regulations and standards that apply to construction projects;
- A3.** use construction terminology correctly;
- A4.** apply mathematical skills and scientific concepts in the planning and building of a variety of construction projects.

SPECIFIC EXPECTATIONS

A1. Materials, Components, and Processes

By the end of this course, students will:

- A1.1** identify building materials used in residential construction (*e.g., natural wood, plywood, engineered wood products such as oriented-strand board [OSB], concrete, brick, adhesives, steel, copper, acrylonitrile butadiene styrene [ABS], cross-linked polyethylene [PEX or XLPE], armoured and non-metallic sheathed electrical cable*);
- A1.2** describe the properties (*e.g., physical, mechanical, thermal*) of natural and manufactured building materials (*e.g., wood studs, steel studs, laminated beams, engineered floor joists*);
- A1.3** identify the components of the various systems in a building (*e.g., structural: foundation, load-bearing wall, trusses, rafters; building envelope: air barrier, insulation, cladding or siding; electrical: service panel, conduit, cable, device boxes, fixtures*);
- A1.4** identify the materials and methods used to manufacture various construction components (*e.g., wood studs, baseboard, moulding, steel beams, plastic pipe, armoured cable, windows, doors, paint*);
- A1.5** identify various construction processes (*e.g., framing, casting concrete in place, bricklaying, tiling, installing millwork, insulating*).

A2. Codes, Regulations, and Standards

By the end of this course, students will:

- A2.1** describe the scope and purpose of the regulations (*e.g., for framing members, built-up beams, electrical wiring, potable water piping, and drain, waste, and vent systems*) in the codes that apply to building construction (*e.g., Ontario Building Code, Ontario Electrical Safety Code, Ontario Fire Code*);
- A2.2** identify other regulations that apply to the design of residential buildings (*e.g., municipal by-laws, zoning, official plans, requirements for barrier-free access*), and describe the process for obtaining construction permits;
- A2.3** identify safety standards that apply to construction projects (*e.g., standards from the Occupational Health and Safety Act, the Canadian Standards Association [CSA], and the Technical Standards and Safety Authority*).

A3. Terminology

By the end of this course, students will:

- A3.1** use correct terminology to identify and describe construction materials (*e.g., concrete forms, studs, lintels, ground-fault circuit interrupters, P-traps, plenums, medium-density fibreboard, plywood, cement, mortar*);

A3.2 use correct terminology to describe construction processes and techniques (*e.g., processes: framing, levelling, squaring, making plumb; techniques: toe-nailing, shimming*);

A3.3 use correct terminology to identify and describe construction tools and equipment (*e.g., tools: builder's level, framing hammer, wire stripper, pliers, tri-square, trowel, pipe cutter, hand saw, reciprocating saw, masonry saw, circular saw, drill; equipment: air compressor, scaffolding, cement mixer, generator, electrical test meter*);

A3.4 use correct terminology to identify and describe a variety of low-rise residential building styles (*e.g., bungalow, two-storey, townhouse, side-split*).

A4. Mathematical Skills and Scientific Concepts

By the end of this course, students will:

A4.1 convert between fractions and decimals, and between imperial and metric units (*e.g., between feet and metres, between Fahrenheit and Celsius temperatures*), using appropriate tables or charts;

A4.2 accurately calculate the layout and materials required for construction tasks (*e.g., layout of open stair stringers, sizes of lintels, loads and wire gauges for electrical circuits, lengths and diameters of drain and vent pipes, duct sizes and lengths for residential heating/cooling systems*);

A4.3 estimate the quantities and costs of materials for construction projects, using appropriate metric and imperial units (*e.g., metres, square metres, cubic metres, board feet, linear feet, square feet, cubic yards*), and estimate the labour costs for these projects;

A4.4 determine lengths and diameters of fastening devices needed to assemble various construction projects (*e.g., lengths and gauges of screws, nails, and staples; diameters of dowels*), using appropriate metric and/or imperial units;

A4.5 demonstrate an understanding of scientific concepts (*e.g., relationships among voltage, current, resistance, and power, and between wire size and current capacity; simple electron theory; methods of heat transfer*) related to construction technology.

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process and/or other problem-solving processes and techniques as appropriate when planning a variety of residential construction projects, and demonstrate an understanding of factors that affect construction design;
- B2.** create and use working drawings for a variety of construction projects;
- B3.** determine the requirements in building codes, regulations, and standards that apply to construction projects, and describe the permit and inspection process;
- B4.** plan the installation of the systems for a building.

SPECIFIC EXPECTATIONS

B1. Design and Planning

By the end of this course, students will:

- B1.1** apply a design process and/or other problem-solving techniques (see pp. 21–23) to address construction technology challenges;
- B1.2** use models, prototypes, and/or sketches to aid in improving the design for a project;
- B1.3** apply design principles (*e.g., scale, proportion, contrast*) when designing residential construction projects (*e.g., kitchen layout, electrical layout, plumbing layout, mechanical layout, structural member layout*);
- B1.4** identify the weather-related loads and stresses that a building must be designed to withstand (*e.g., force of wind, snow load on roofs, expansion and contraction due to changes in temperature and humidity*), and describe how these factors affect the planning and design of a construction project;
- B1.5** provide rationales for using particular materials for construction projects (*e.g., asphalt shingles versus steel roofing, wood studs versus steel studs, block versus poured concrete, natural wood joists versus engineered joists*).

B2. Working Drawings

By the end of this course, students will:

- B2.1** create sketches and/or technical drawings (*e.g., orthographic, isometric, plan, elevation,*

section, detail) of one or more building systems (*e.g., structural, plumbing, electrical*), using manual and/or computer-assisted methods and appropriate metric and/or imperial units;

- B2.2** use working drawings to help plan the sequence of tasks for completing a construction project (*e.g., excavation, footings, foundation walls, framing, sheathing, rough-in of mechanical systems, drywalling, installation of devices and fixtures*);
- B2.3** interpret working drawings to accurately lay out projects (*e.g., determine dimensions and placement of footings, columns, openings, and beams; determine specified materials, size and type of windows and doors, masonry bond pattern, and locations of electrical, mechanical, and plumbing components*).

B3. Codes, Regulations, and Standards

By the end of this course, students will:

- B3.1** identify the types of loads (*e.g., live, dead*) that building codes require residential structural members (*e.g., footings, foundations, floors, bearing walls, roofs, columns, beams, lintels*) to withstand;
- B3.2** use various resources to determine the requirements in codes, regulations, and standards that apply to construction projects (*e.g., reference charts and tables; published codes, regulations, and standards; guides and trade manuals; government and association websites*);

B3.3 describe the permit and inspection process required for various aspects of construction projects (*e.g., building, electrical, plumbing, HVAC*).

B4. Building Systems

By the end of this course, students will:

B4.1 plan systems for construction projects (*e.g., foundation, framing, roof, electrical, plumbing*);

B4.2 identify the factors affecting foundation design (*e.g., drainage, soils, frost penetration*), and allow for these factors when planning and designing a construction project;

B4.3 identify the relationships between structural and decorative elements in construction projects (*e.g., studs, toe plates, and top plates provide secure support for baseboards and mouldings, which cover joints where walls meet floors and ceilings; masonry bond patterns provide strength and an attractive appearance*) and incorporate these relationships into the planning and design of construction projects;

B4.4 recommend modifications to a building project that would improve its quality and value (*e.g., upgrades to insulation, air barrier, or HVAC system to reduce operating costs; use of more attractive or durable fixtures, trim, flooring, or exterior finish*).

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate appropriate technical skills, including the safe use of construction tools, equipment, and materials;
- C2.** demonstrate safe and accurate techniques for assembling construction projects;
- C3.** apply various finishes to complete construction projects.

SPECIFIC EXPECTATIONS

C1. Technical Skills

By the end of this course, students will:

- C1.1** use, maintain, and store construction tools, equipment, and materials safely and correctly (*e.g., tools: builder's level, framing hammer, wire stripper, pliers, tri-square, trowel, pipe cutter, hand saw, reciprocating saw, masonry saw, circular saw, drill; equipment: air compressor, scaffolding, cement mixer, generator, electrical test meter; materials: lumber, sheet goods, plumbing materials, bricks, wiring*);
- C1.2** lay out construction projects using a variety of tools and equipment (*e.g., framing square, laser level, string line, plumb bob*);
- C1.3** use metric and imperial units correctly during the construction of a variety of projects;
- C1.4** communicate construction information verbally and graphically (*e.g., through written instructions, sketches, and/or detail drawings*).

C2. Building and Assembly

By the end of this course, students will:

- C2.1** construct or install components of projects according to working drawings, lists of materials and fixtures, and other design specifications (*e.g., for roof pitch, masonry bond pattern, plumbing and electrical fixtures, tile patterns, mechanical system components, and electrical and structured wiring*);

- C2.2** install, test, and, if necessary, troubleshoot various systems of a construction project (*e.g., electrical, plumbing, heating, masonry*), ensuring that these systems comply with all applicable codes, regulations, and standards;

- C2.3** use a variety of appropriate techniques for levelling, plumbing, bracing, and squaring construction components;

- C2.4** implement appropriate solutions for construction problems or challenges.

C3. Finishing

By the end of this course, students will:

- C3.1** apply appropriate exterior materials to finish construction projects (*e.g., vinyl or wood siding, fascia, soffits, brick veneer*);
- C3.2** apply appropriate interior materials to finish construction projects (*e.g., paint, tile, trim*).

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of the environmental effects of construction projects, and ways of reducing harmful effects;
- D2.** demonstrate an understanding of how society and the construction industry affect each other.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** compare the efficiency and environmental effects of a variety of energy sources used in residential dwellings (*e.g., solar, ground source, pellets, propane, wood, oil, natural gas, wind*);
- D1.2** plan projects and processes to minimize waste (*e.g., use efficient cutting patterns, reuse and recycle leftover materials*);
- D1.3** identify ways of reducing energy consumption in the home (*e.g., heat recovery ventilator, tankless water heater, timers, energy-efficient lighting*);
- D1.4** identify programs provided by community partners and government agencies to reduce construction waste and to reuse or recycle construction materials (*e.g., drywall, wood, refrigerants*);
- D1.5** research and identify sources and certifications for construction materials that have been manufactured using sustainable practices.

D2. Technology and Society

By the end of this course, students will:

- D2.1** identify how societal and client needs (*e.g., budget constraints, barrier-free access, energy efficiency, cultural preferences*) affect construction projects;
- D2.2** describe how features of residential buildings (*e.g., halls, atriums, stairs, ramps*) can affect the occupants' quality of life;
- D2.3** identify the economic and social effects of the construction industry on a community or region (*e.g., direct and indirect effects on employment, waste disposal, land use, water supply, Aboriginal land claims and traditional hunting*).

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of and comply with health and safety regulations and practices specific to the construction industry;
- E2.** describe career opportunities in the construction industry, and explain the importance of lifelong learning in this field.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** identify hazards related to construction materials, processes, tools, and equipment (*e.g., toxic or flammable fumes from solvents, paints, varnishes, and gasoline; explosion or burns from propane; lung damage from silica; tripping or falls in unfinished buildings; shock from damaged power tools or electrical equipment*), and the precautions that should be taken to avoid these hazards;
- E1.2** identify and comply with health and safety legislation and practices for the construction industry (*e.g., Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, Ontario Building Code, local by-laws*);
- E1.3** use, handle, and store materials in accordance with Workplace Hazardous Materials Information System (WHMIS) guidelines;
- E1.4** describe the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.5** use protective clothing, gear, and equipment appropriately (*e.g. dust mask, safety glasses, safety harness*).

E2. Career Opportunities

By the end of this course, students will:

- E2.1** identify careers in the construction industry (*e.g., construction craft worker, labourer, tradesperson, estimator, entrepreneur, interior designer,*

artisan), and describe the education and training required for these careers;

- E2.2** explain the importance of lifelong learning for someone choosing a career in the construction industry;
- E2.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the construction industry (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- E2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., finding information, job task planning and organizing, measurement and calculation*);
- E2.5** demonstrate an understanding of and apply the work habits that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability*);
- E2.6** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in construction technology (*e.g., Passport to Safety certificate, technical drawings, reports, photographs of projects, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.